

"Student Activity Tracker App"

A small React app where students can:

- Type a student's name (useRef)
- Add student using button (useCallback)
- All students show in a list (props)
- Automatically update total student count (useEffect)
- Keep track using state (useState)

This project makes all concepts work together.

★ COMPONENT STRUCTURE

```
App.jsx
├── StudentInput.jsx
├── StudentList.jsx
└── StudentCard.jsx
```

★ FEATURES (What student must build)

1 useRef

Focus input automatically when page loads

Clear input using ref

Read input value without re-rendering

2 useState

Store list of students

Store the total student count

Store input changes

3 useCallback

Create a memoized function to add a student

So no new function is created on re-renders

4 useEffect

Whenever student list changes → update total count
Log when component mounts and updates

5 Props

App → StudentList (props)

StudentList → StudentCard (props)

★ TASK REQUIREMENT FOR STUDENTS

Students must:

- ✓ Complete all React components
- ✓ Add console logs to see hook behavior
- ✓ Understand how each hook connects to the next
- ✓ Provide explanation of flow

★ COMPLETE FLOW DIAGRAM

```
(1) StudentInput
    |   useRef reads input
    ↓
(2) addStudent callback (useCallback)
    ↓
(3) App updates students state (useState)
    ↓
(4) useEffect logs update
    ↓
(5) StudentList gets updated students (props)
    ↓
(6) StudentCard displays each student (props)
```